



SATKURI SUMITH CHANDRA

B.Tech.(Hons.) in COMPUTER SCIENCE AND ENGINEERING at
IIT KHARAGPUR

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🎓 EDUCATION

IIT Kharagpur, B.TECH
2022 – 2026 | India
• CGPA: 8.74 / 10

🏆 ACHIEVEMENTS

JEE Main 2022, NTA
• All India Rank 1329

JEE Advanced 2022, IIT
• All India Rank 883

National Talent Search Examination
(NTSE) Scholarship, NCERT

★ COMPETITION

Codeforces,
Candidate Master (1923 rating)
• Handle: Sumith_chandra

🧠 SKILLS AND EXPERTISE

Programming Languages: C | C++ |
Python | Golang | SQL

Tools and Software: Git | Github |
Ollama | Linux

AI/ML Frameworks &
Libraries: PyTorch | Transformers |
LangChain | LangGraph |
DeepSpeed | HuggingFace |
ChromaDB | NumPy | Pandas |
Agentic AI | API | LLMs

📖 COURSEWORK INFORMATION

Relevant Coursework, IIT Kharagpur
Machine Learning, Deep Learning,
Information Retrieval, High
Performance Parallel Programming,
Operating Systems, Computer
Networks, Database Management
Systems.

📁 INTERNSHIPS

Fine-tuning LLMs Under Resource Constraints, Samsung R&D Institute
05/2025 – 07/2025 | Bangalore, India

- **Reduced** LLM model **size by 50%** and **accelerated inference by 2x** by engineering PTQ and AWQ quantization pipelines.
- **Cut GPU memory usage by 60%** during fine-tuning by applying LoRA and QLoRA parameter-efficient adaptation techniques.
- **Enabled 4x longer sequence training** within fixed memory budgets by integrating DeepSpeed ZeRO parallelism and Liger-kernel.
- Delivered a production-ready on-device LLM retaining **95%+ task accuracy** by combining quantization, PEFT, and distributed training.

📁 PROJECTS

Defending Deep Learning Models against Adversarial Attacks,
B.Tech Project
08/2025 – 04/2026

- **Restored** VideoMAE **robust accuracy from 0% to 85%+** under PGD attacks by building a dynamic defense pipeline with attack-type routing.
- Maintained **94%+ clean accuracy** across **PGD, OFA, and C&W attacks** by designing a dual-detector with **ResNet-18** optical flow and Integrated Gradients ensemble.
- **Neutralized adversarial perturbations** by implementing intelligent spatial transformation routing with adaptive resizing and cropping.

PaperMind — AI-Powered RAG Agent with Agentic Workflow
04/2026 – 05/2026

- **Built an agentic query-routing system** using a LangGraph state machine, **automating intent classification across 3 routes** (document RAG, web search, direct answer) — eliminating manual query handling.
- **Achieved page-cited document QA with zero hallucination on weak-context queries** by implementing a ChromaDB RAG pipeline with cosine-similarity-based smart fallback to Tavily web search.
- **Delivered real-time streaming chat UI with multi-turn memory** by building a FastAPI + SSE backend serving a React 19 frontend, using a dual-LLM architecture (Llama 3.1 8B routing + Llama 3.3 70B generation) on Groq.

Image Captioning using Transformers, Course Project: Deep Learning
03/2025 – 04/2025

- **Improved** captioning **BLEU score by 26%** over SmolVLM baseline (0.0421 vs 0.0334) by training a custom ViT-GPT2 model.
- **Validated robustness** under degraded inputs by evaluating semantic accuracy across progressive image occlusion levels.
- **Achieved 97% model attribution accuracy** by deploying a fine-tuned BERT classifier to distinguish custom vs. baseline outputs.